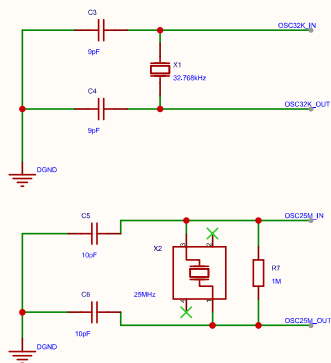
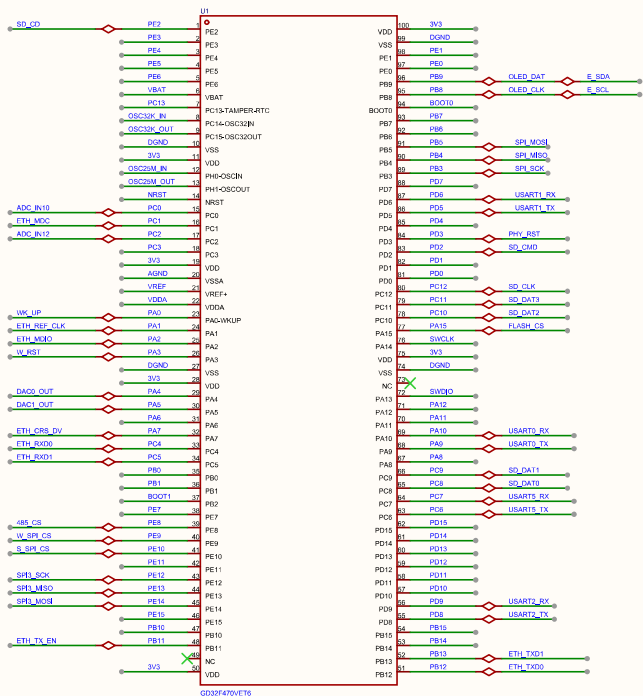
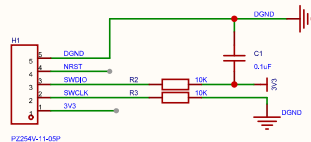


去耦电容

每个供电引脚旁边放俩

MCU
GD32F470VET6

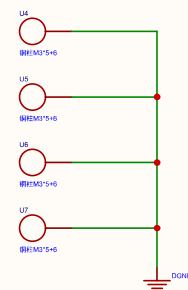
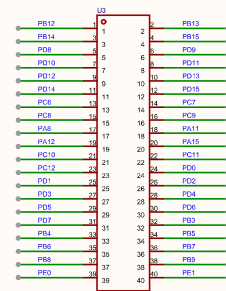


SWDIO上拉电阻：抑制噪声，防止随机翻转，提升可靠性。

SWCLK下拉电阻：防止浮空导致的噪声，提升可靠性。

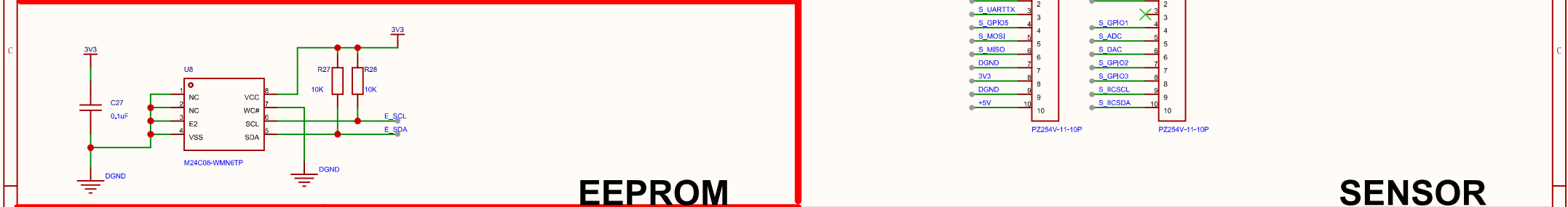
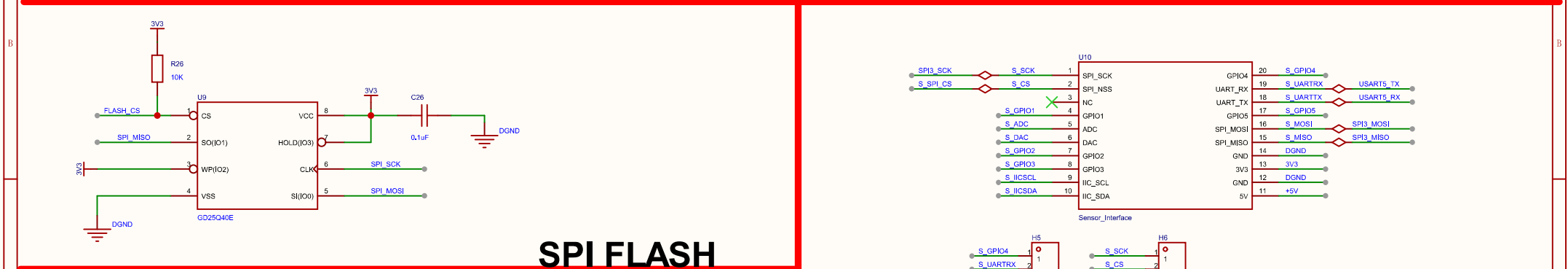
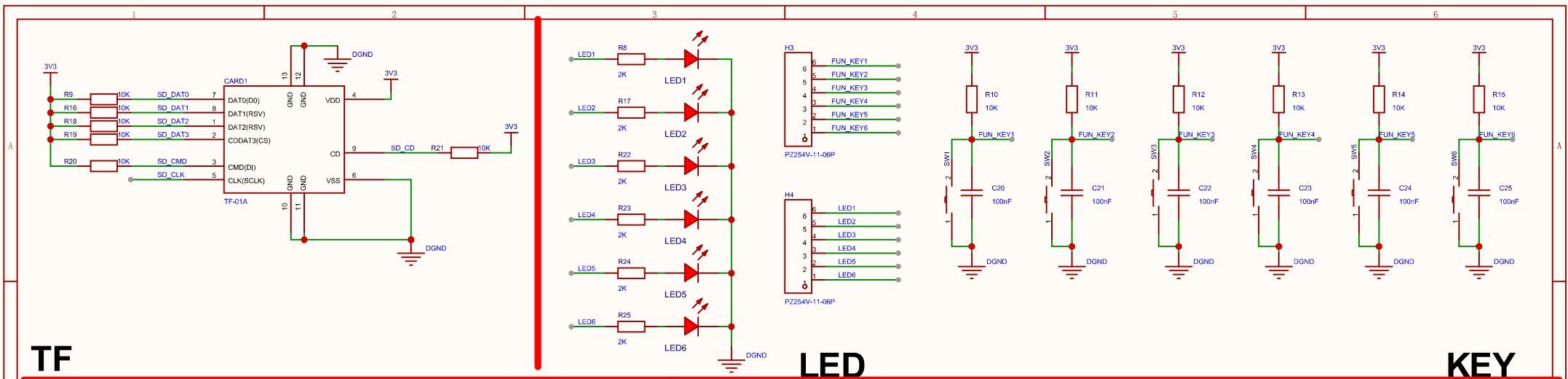


BOOT

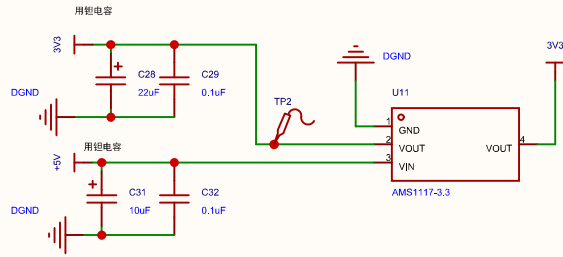
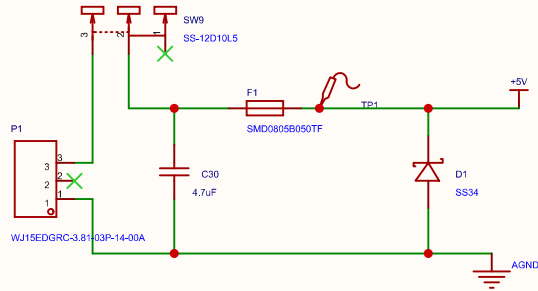


Pin Out

原理图	CoreBoard		创建日期	2026-02-06
			更新日期	2026-04-07
板子	CIMC_GD32F4_Development_Kit		图页	MCU_Mainboard
绘制	Lingyu Meng	GD32F470 Development Kit V2.0		
审阅	Lingyu Meng			
		版本	尺寸	页 1 共 5
 中国智能制造挑战赛 China Intelligent Manufacturing Challenge		V1.0	A4	CIMC“西门子杯”智能制造挑战赛

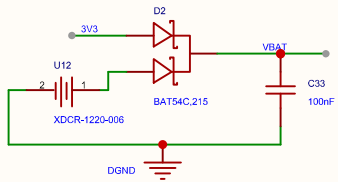


5V输入

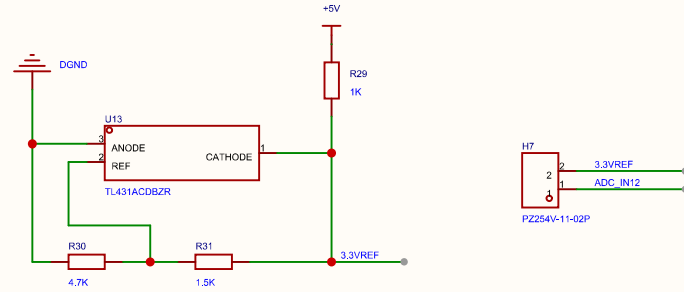


5V - 3V3

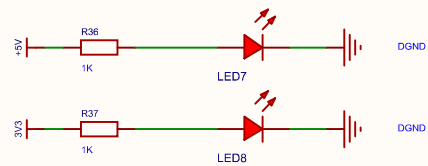
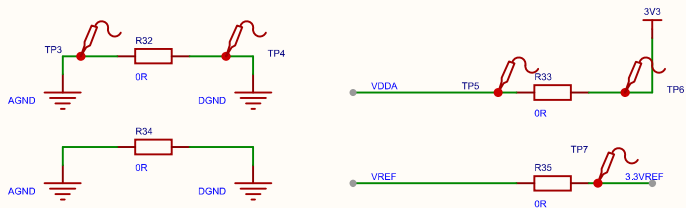
Vbat



$$V_{REF} = 2.5 * (1 + R1(1.5k)/R2(4.7k)) = 3.2979V$$



ADC基准



5V 电源灯



3.3V 电源灯

原理图	CoreBoard		创建日期	2026-02-06
板子	CIMC_GD32F4_Development_Kit		更新日期	2026-04-07
绘制	Lingyu Meng	GD32F470 Development Kit V2.0		
审阅	Lingyu Meng			
		版本	尺寸	页 3 共 5
		V1.0	A4	CIMC “西门子杯” 智能制造挑战赛

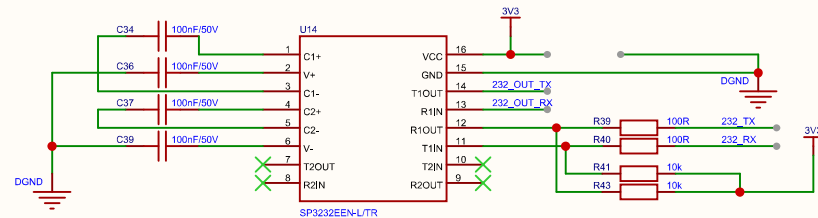
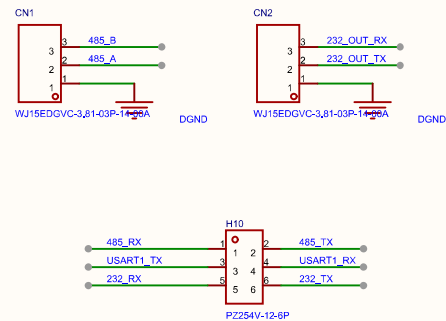
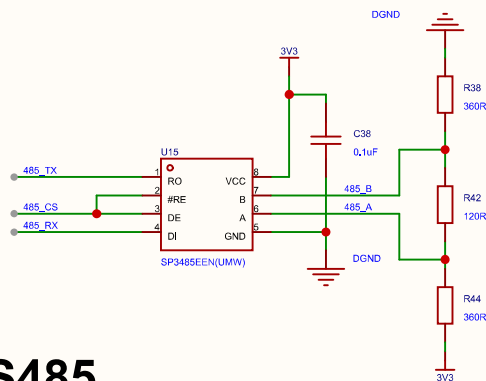
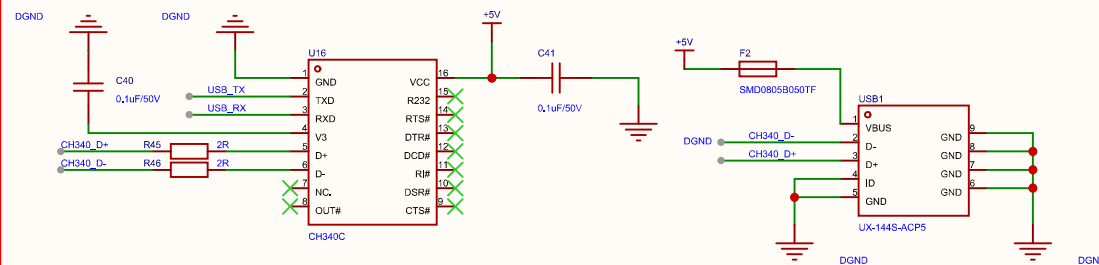


Diagram illustrating the pin connections for the ESP8266 module:

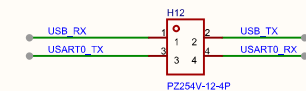
- Header CN3 (WJ15EDGVC-3,8-03P-140A):**
 - Pin 3: TTL_OUT_RX
 - Pin 2: TTL_OUT_TX
 - Pin 1: GND
- Header H11 (P2254V-12-6P):**
 - Pin 1: GND
 - Pin 2: TTL_OUT_RX
 - Pin 3: USART2_TX
 - Pin 4: USART2_RX
 - Pin 5: ESP_RX
 - Pin 6: ESP_TX

USART2

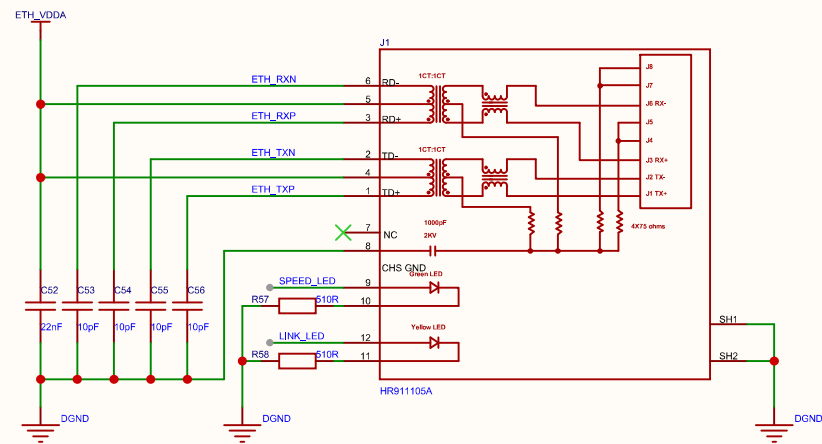
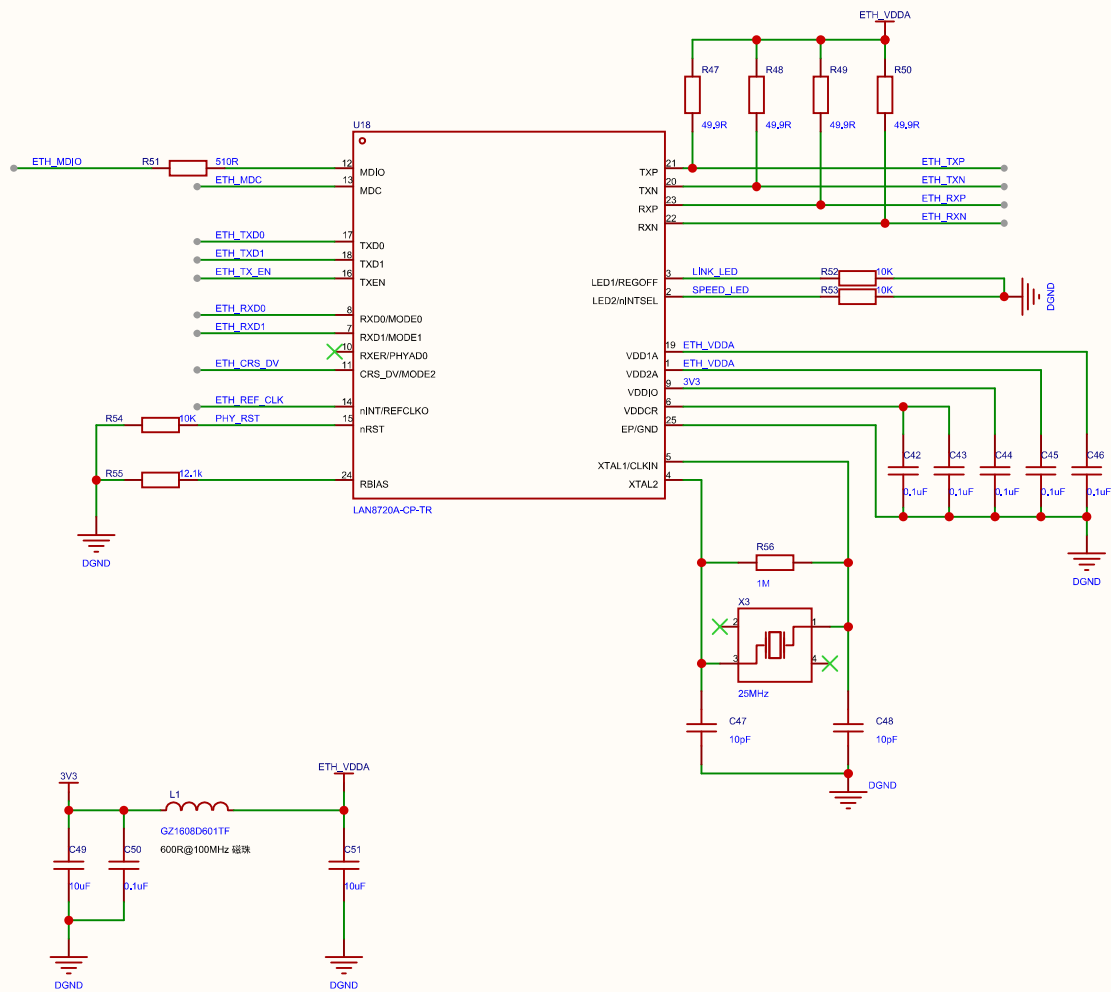


USB

USART0



中国智能制造挑战赛
China Intelligent Manufacturing Challenge



原理图	CoreBoard		创建日期	2026-02-06
			更新日期	2026-02-06
板子	CIMC_GD32F4_Development_Kit		图页	Ethernet
绘制	GD32F470 Development Kit V2.0			
审阅				
		版本	尺寸	页 5 共 5
		V1.0	A4	嘉立创EDA